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RESEARCH ARTICLE

# Short-Term Load Forecasting for Electrical Power Distribution Systems Using Enhanced Deep Neural Networks

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**ABSTRACT** The rationale for using enhanced Deep Neural Networks (DNNs) in the power distribution system for short-term load forecasting (STLF) originates from a thorough analysis of current trends, the emergence of the state-of-the-art use cases and approaches. STLF plays a crucial role in economic load dispatch, hydrothermal coordination, system security assessment, load shedding, unit commitment, and cost-effective risk management in power systems with renewable energy sources. In this study, we introduce a Long Short-Term Memory (LSTM), augmented with enhancements inspired by an Efficient and Parallel Genetic Algorithm (EPGA) for the STLF of Jimma town power distribution system. To forecast the load in the short term, the model takes into account wind direction, wind speed, humidity, temperature, season, load history, and peak load due to holidays. The optimal linear combination of inputs for determining daily load is derived using EPGA and data from Ethiopian Electric Utility (EEU). The linearly combined data is then fed into the LSTM model for load prediction. During training, this allows the LSTM model to focus on the pattern of a single time-series data rather than the best combination of many input patterns. The proposed method uses the combination of EPGA and LSTM models for accurate STLF. Thorough experimental analysis indicates that the root-mean-squared error (RMSE) achieved by the EPGA-enhanced LSTM for Jimma town power distribution system STLF is about 43.87. This represents a 7.486% improvement over the prediction obtained using only LSTM model. Additionally, the mean average percentage error (MAPE) for five sample loads used to test the EPGA-enhanced LSTM prediction method further supports the robustness of the proposed method.

**INDEX TERMS** Deep neural networks, long short-term memory, load forecast, short-term load forecast, efficient and parallel genetic algorithm, EPGA enhanced LSTM.

## I. INTRODUCTION

Balancing demand and supply in the economics of power systems planning and operation has been a moving target for decades [1], [2], [3], [4]. This moving target gets complex and cumbersome with the introduction of intermittent renewables and variable needs for electricity. As this trend will mainly be part of the future renewable energy systems, it makes it a challenge worth facing. Furthermore, failing to continuously balance demand and supply in renewable energy generation leads to operational contingencies that are closely related to unnecessary loadings [5], [6], [7].

Operational contingencies put the considered power system's reliability, stability and security at risk. In addition to operational contingencies, imposed costs, the inability to properly manage the continuous balance and service of the energy market sabotages its economic profits [8].

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Therefore, the occurrence of blackouts and economic motivations are the main reasons for the growing quest to address the gap between electricity demand and supply [7]. Moreover, the occurrence of faults due to unnecessary loading can be avoided by proper operational management of the load demands [9].

Basically, the layout for supply depends on the knowledge of all the attributes of the demand. This is to mean that, determining demand comes before determining supply. These operational contingencies and demand-supply imbalances are the main causes of frequent electricity blackouts in the most developing countries including the Jimma distribution system in Ethiopia.

In line with these motivations, the storage constraint of electrical energy adds weight to the importance of estimating the load demand in advance as electrical energy has to be generated when there is demand for it. In practical terms, the estimation of load in advance is called load forecasting. The importance of load forecasting in addressing this gap thus hinges on the relationship between operational contingencies, storage constraints and demand-supply imbalance.

Load forecasting is the process of predicting and estimating future electrical loads in advance and identifying ways in which they can be met. It is vital for an electric industry in a deregulated energy market. It also benefits the energy system operator in economic load dispatch, hydrothermal coordination, system security assessment, load shedding, unit commitment and cost-effective risk management. From the time horizon perspective, load forecasting is categorized into three i.e. Short Term Load Forecasting (STLF) [5], [10], [11], [12], Medium Term Load Forecasting (MTLF) [5], [13], [14] and Long Term Load Forecasting (LTLF) [9]. STLF ranging from day ahead hours to weeks is a computationally difficult optimization problem. Mathematically modelling the probability distribution function of the stochastic nature of the load and predicting it in a day ahead manner is also a plus to the moving target of balancing load and supply.

In this study, STLF that employs enhanced Deep Neural Networks (DNNs) to quantify the uncertainty in the future load of Jimma city electric power distribution system by developing accurate, reliable and stable model that specify the relationship between load and its key drivers is presented. An enhanced, continuous, multilayered Recurrent Neural Network (RNN) called Long Short-Term Memory (LSTM) combined with Efficient and Parallel Genetic Algorithm (EPGA) STLF method is used in this work. In this proposed method, the historical load data and weather parameters that affect energy consumption are first linearly combined using EPGA. Then, the linearly combined data is fed to the LSTM model for STLF of Jimma town electric power distribution system. This reduces the parameters of the LSTM model as it is fed with a single time series data for load forecasting.

The main contribution of this study is the introduction of EPGA enhanced LSTM for Jimma city electric power distribution system's load prediction in a short-term interval considering weather load model and demand profiles affected by distributed generators like Solar Home Systems and diesel generators.

The remaining section of this paper are outlined as follows. In Section II, a comprehensive literature review is provided. Data collection and analysis is presented in Section III. The proposed method is detailed in Section IV. Finally, concluding remarks and recommendations for future work are provided in Section V.

## II. LITERATURE REVIEW

The motivation for developing enhanced DNN for forecasting the Jimma city electric power distribution system's load in a short-term interval comes from a comprehensive analysis of the recent trends, state-of-the-art applications and approaches of STLF with a special focus on enhanced DNN. In this regard, recent research works on deep learning-based short-term electricity load forecasts are presented in this section.

Traditional modelling methods possess computational patterns that make them inefficient to capture long-term dependencies of multivariate data with high prediction accuracy [1], [7], [13], [15], [16], [17], [18], [19], and [20]. To address these issues, diverse combinations of recurrent neural network (RNN) and convolutional neural network (CNN) and enhancements of other deep learning models were proposed [1], [5], [12], [19], [21], [22], [23]. Ioannis Patsakos [24] presented a survey on the recent development of deep learning methods applied to STLF in addition to the review on STLF models for renewable micro-grids applications in [2], [25], [26], [27], and [28].

Butt et al. [9] combined Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) to capture the temporal correlation in the time series data in order to improve forecasting performance. A new DNN framework that integrates the CNN model and the LSTM model was proposed to improve the forecasting accuracy in [29]. CNN was also enhanced with the introduction of the Time Dependent (TD) factor to it [30], [31]. These combined models' detailed experiments were tested in a physical case study. The findings of this study show that the developed structure can be used in local energy system planning and dispatch, scheduling and maintenance operations.

Memory and Perceptron [32] introduced a forecasting model called a two-stage STLF model combining LSTM and multilayer perceptron (MLP) to lower forecast errors, operating costs and risks. Bian et al. [33] introduced a hybrid deep learning model that takes into account the accumulated effect of temperature called Temporal Convolutional Network (TCN)- Deep Neural Network (DNN). This hybrid scheme achieved 97.92% accuracy in forecasting.

TCN combined with Variation Mode Decomposition (VMD) was promisingly employed in error correction of STLF [34]. This prediction accuracy was increased to 98.94% in [21], [30], [35], [36], [37], [38], [39], [40], [41], and [42] after combining Convolutional Neural Network (CNN) and Long- Term Short- Term Memory recurrent neural network

**TABLE 1.** Summarized review of the existing deep learning models and their performances for STLF.

| Ref. | DNN models | Test system | MAPE | RMSE |
|---|---|---|---|---|
| [22] | CNN-GRU based hybrid approach Hybrid sequential learning | Individual Household Electric Power Consumption (IHEPC) datasets | 0.24 | 0.31 |
| [30] | a CNN-LSTM network with multiple heads (MCNN-LSTM) | Ireland dataset | 2.93 | 53.73 |
| [58] | Genetic Algorithm determined DFNN | Northavon House in University of the West of England, Bristol | 0.0987 | 55.25 |
| [50] | Feature Extraction and GA-enhanced adaptive DNN | Northavon House in University of the West of England, Bristol | 0.071 | 37.7 |
| [5] | STLF-Net: Two-stream DNN Gated Recurrent Units (GRUs) and Temporal Convolutional (TC) | residential buildings (IHEPC and AEP) | 36.4 | 0.4386 |
| [14] | LSTM and RNN Deep Neural Network | FESCO, a company in Pakistan | 1.09 | - |
| [62] | optimally pruned extreme learning machines (OP-ELM), and a deep learning technique, long short-term memory (LSTM) | South African power utility | 0.1384 | 0.0717 |
| [40] | Graph Convolutional Network (GCN) and Long Short-Term Memory Network (LSTM). | PJM, United States | 7.50 | - |
| [29] | CNN and LSTM | Question B of the 10th Teddy Cup Data Mining Challenge | 6.62 | 2.72 |
| [20] | Parallel Deep LSTM-CNN | Load consumption of Malaysia and Germany | 8.81 | 0.36 |
| [43] | hybrid CNN-LSTM model | Public data of the Smart Grid Smart City (SGSC) project | 44.06 | 4.76 |
| [54] | GLSTM model: Genetic LSTM model (LSTM and GA) | European Center for Medium-range Weather Forecasts (ECMWF) | 0.01864 | 0.1365 |
| [55], [58] | LSTM network, and Genetic algorithm (GA) | New England (ISO-NE) | 0.6710 | - |

(LSTM). Hybrid CNN-LSTM-based STLF was applied for the residential level load forecasting in [43].

LSTM was also hybridized with Backpropagation Neural Networks in [44], Neural Prophet (NP) and RNN [45]. Another study achieved prediction accuracy of 97.52%–98.92% by introducing a self-adaptive deep learning model [5], [28], [46], [47].

A review of DNN models employed for building-level load forecasting is presented in [7]. Hybrid deep learning-based predictions were used for physical case studies such as Saudi Smart Grid [1], Australia [11], [48] Bangladesh [49] and the United Kingdom [24], [46], [50], [51]. Genetic Algorithms were recently employed to enhance and improve the prediction accuracy of DNNs [1], [14], [21], [24], [50], [51], and [52].

These trends showcased an improvement in capturing the intermittency of weather-dependent STLF of distribution systems fed by renewable energy systems. Optimizing LSTM networks using genetic algorithms has recently become a new trend in an attempt to exploit the global optimization capability of genetic algorithms [5], [53], [54], [55].

Recent studies [8], [11], [18], [36], [37], [45], [57], and [58] deciphered that the difficulty of training DNNs is one of the prominent computational challenges in this domain. Deep Neural networks based short-term energy demand forecast are summarized and presented in Table 1.

In addition to the above studies, there has been substantial advancement in applying forecasting methods that take advantage of combining and hybridizing features of novel methods [9], [12], [25], [30], [34], [42], [44], [57], [64], [65], and [66]. However, these advancements are impeded by the forecasting difficulty bestowed by the introduction of renewable generation and unpredictable weather conditions [21], [67]. Generally, underlining the strengths, this literature review paves a way to articulate the following research gaps in this research domain:

- There is a lack of research on recent advances and improvements of hybrid and enhanced deep learning methods applied to short-term forecasting at a distribution system level.
- There is a lack of DNN-based research and data when it comes to short-term load forecasting of the Ethiopian power grid, distribution systems and sector-based demand profiles.

The proposed method combines the strengths of EPGA and LSTM to implement a robust STLF deep learning model for Jimma city (Ethiopia) electric power distribution system. In our model, the EPGA is used to linearly combine the weather data and historical load data to generate a single time series data. This data is then fed to LSTM model for STLF. The EPGA is used to here to simplify the complexity of LSTM and improve the robustness of STLF.

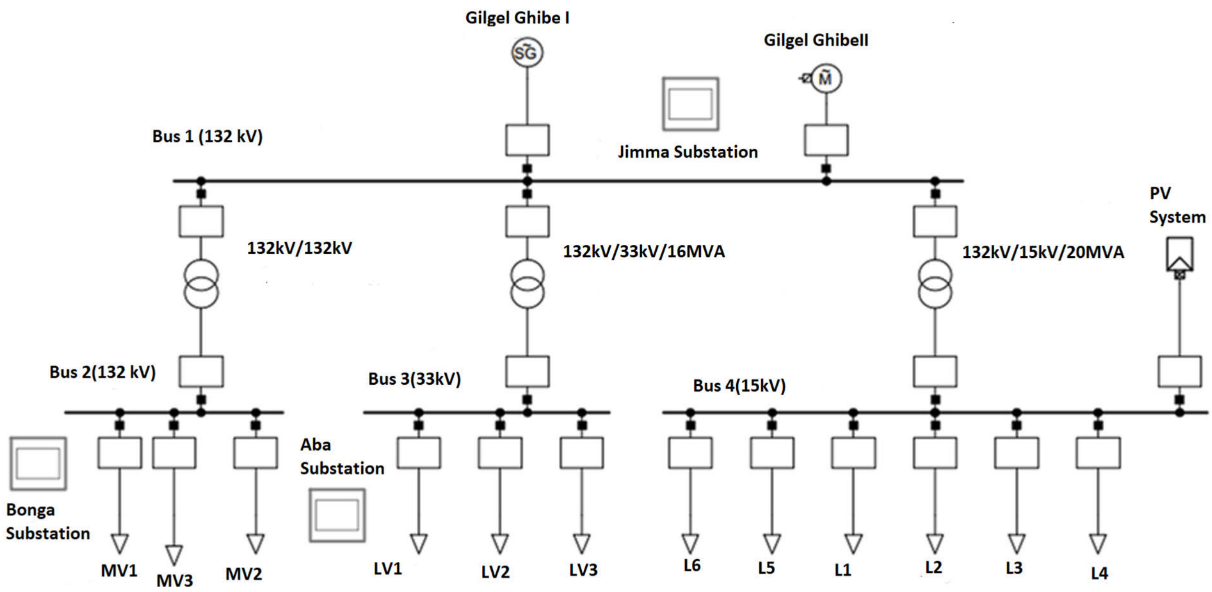


**FIGURE 1.** One-line diagram of the current Jimma electrical distribution system incorporating distributed generators.

## III. DATA COLLECTION AND ANALYSIS

Enhanced deep neural networks-based short-term electrical load forecast can significantly improve power system planning and provides information on expected load growth, load profiles, economic dispatch and load distribution; helping to subsequently reduce the number of service interruptions and improving the system reliability. As deep neural networks are data intensive computational models, set of data such as historical load data, weather data, weather forecast data, and real-time database that includes Gross Domestic Product (GDP) and population are collected from NASA Surface Meteorology and Solar Energy, Load dispatch centre in Addis Ababa, Ethiopian Electric Utility (EEU), and Jimma city electric distribution system. The data has been used to train and validate the proposed STLF model for Jimma city distribution system. Jimma is the largest city in southwestern Ethiopia. It is a special zone of the Oromia Region and is surrounded by the Jimma Zone. It has a latitude and longitude of 7°40′ N 36°50′ E. The town was the capital of Kaffa Province until the province was dissolved. Prior to the 2007 census, Jimma was re-organized administratively as a special Zone [68]. Before 1989 the town got electricity from diesel generators. The town's substation started its function in 1989 with a 6.3 MVA transformer rating. At that time the peak load consumption was 1.5 MW. In 2005, the substation was upgraded to 30 MVA which is working up to now. Nowadays the town utilizes a maximum of around 25 MW of power [69]. Jimma city electric distribution system is now being fed from Gilgel Gibe I and II hydropower generation stations. As a result, it can be referred to as four bus distribution system with distributed Solar Home Systems (SHS) and scattered diesel generators. It also transfers power to two other substations called Bonga substation and the Aba Substation. The distribution system is comprised of 12 feeders and 4 busses as depicted in Figure 1. Five hundred distribution lines connecting the 4 buses and 12 feeders supply the distribution system's customers with electricity.

Load forecasting depends on the Gross Domestic Product (GDP) of the site under consideration, population (pop), energy demand ($P_D$), GDP per capita (GDP/cap), energy loss ($P_L$), load factor (LF), levelized cost of energy (LOCE) and the service time of the load. Mathematically, the key drivers of accurate load forecast can be represented by [11] and [70]:

$$P_F = a_0 + a_1 GDP + a_2 Pop + a_3 P_D + a_4 (GDP/Cap) + a_5 P_L + a_6 LF + a_7 LCOE + a_8 time \quad (1)$$

where $P_F$ is the forecasted load, and $a_i$ is the weight of each factor on the load prediction task from 1 to 8. Practically speaking, GDP, GDP per capita, population, load factor and LOCE are almost constant daily. This means these factors can only affect either MTLF or LTLF. Therefore, a better mathematical model to represent the effect of key deriving factors for STLF is given by:

$$P_F = a_0 + a_1 w_{direction} + a_2 A_{pressure} + a_3 P_D + a_4 (T) + a_5 P_L + a_6 v_w + a_7 (Humudity) + a_8 time \quad (2)$$

where: $w_{direction}$, $A_{pressure}$, $T$, $v_w$ and denote the direction of wind, surface pressure, temperature and wind speed, respectively. Temperature in this context refers to the effect of dew temperature, wet bulb temperature, earth skin temperature and atmospheric temperature. Then, the weather load model ($L_{weather}$), equation (3), follows with its distinct parameters [50];

$$L_{weather} = k_w \left(T_w - x\sigma_T + T'\right) T \leq T_w \quad (3)$$

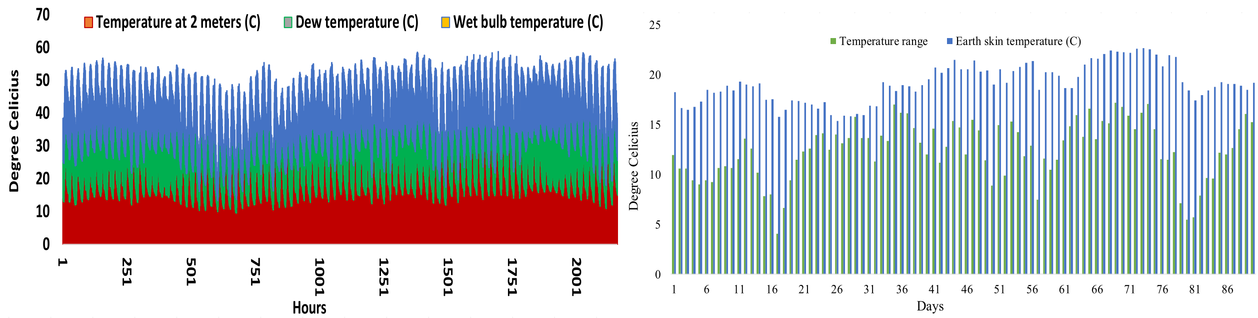


**FIGURE 2.** Temperature, dew temperature, wet bulb temperature, earth skin temperature and temperature range of jimma town, february 2023.

where: $k_w$ is the slope of the temperature thresholds, in the probability of finding the optimal weather-sensitive minimum temperature range $\sigma_T$ and maximum temperature $T'$.

In accordance with the mathematically formulated STLF model in (2), relevant data such as dew temperature, wet bulb temperature, earth skin temperature, temperature range wind speed, surface pressure, wind direction and specific humidity of Jimma town were collected from NASA Meteorology and Solar Energy (SSE) data and depicted in Figures 2 and 3.

Additionally, historical energy consumption data presented in Table 2 were considered in the STLF model formulation and prediction. Furthermore, real-time invoices of energy consumption, Tables 2 and 3, were collected for meta-analysis of relevant energy consumption data summarized in Figure 4. The weather and historical load data collected pertinent to Jimma city distribution system (Figure 2, Figure 3 and Figure 4) show that a highly nonlinear prediction model is required for accurate STLF.

Once the data shown in the above figures and tables are collected and normalized, and then a linear combination of the weather data and historical load data was carried out using EPGA. Then, the LSTM model is trained using the combined data for STLF of Jimma city distribution system.

## IV. SHORT-TERM LOAD FORECASTING USING EPGA-ENHANCED LSTM

When artificial neural networks (ANN)-based energy demand forecasting is considered, its computational performance depends on how the input layers, hidden layers, and output layers are configured [52]. Deep Neural Network (DNN) is a particular Multi-Layer Perceptron (MLP) neural network having more than two hidden layers [6], [11], [16].

Typically, the increase in the number of hidden layers improves DNN's computational performance. However, increasing the number of hidden layers can cause an over-fitting problem during training of the model. This subsequently leads to another challenge, called the vanishing gradient of an activation function [52].

One way to overcome the overfitting problem is to reduce parameters or size of the deep learning model. In this work, the number of parameters of LSTM model has been reduced by introducing efficient parallel genetic algorithms (EPGA) for optimally combining the weather and historical load data. EPGA is a population-based adaptive optimization method derived from the Darwinian notion of natural selection. It is used to determine the coefficients ($a_i$, i=1…, 8) in equation (2). This reduces the size of the input data to be fed to LSTM model from eight to a single value.

LSTM is an augmented recurrent neural network model. It learns sequential information with long-term dependencies and preserves information for a long period. Traditional recurrent neural network suffers from the vanishing gradient problem [71]. That is, as the number of layers using the same activation function increases, the gradients of the loss function approach zero, making it difficult to train the network through backpropagation of errors.

To prevent the vanishing gradient problem, LSTM architecture deploys memory blocks to overcome the vanishing and exploding gradient problems rather than the conventional RNNs unit that employs simple neural networks. An LSTM architecture [47], [72], [73], [74], [75] involves three gates, which are the input gate, the forget gate, and the output gate.

LSTM utilizes memory cells, as depicted Figure 5, where each cell maintains a cell state and a hidden cell state and uses three gates (namely, input gate, output gate, and forget gate) to control the flow of information into or out of the cell. The gate uses a sigmoid activation function, while input and cell state are usually transformed using tanh function.

The input ($i_t$) of a cell at the time step t of the LSTM can be determined using the following equation.

$$i_t = \sigma \left( W_{ix} x_t + W_{ih} h_{t-1} + W_{ic} c_{t-1} + b_i \right) \tag{4}$$

where;

$W_{ix}$, $W_{ih}$, $W_{ic}$ are incoming links and $b_i$ are bias inputs of the current input $x_t$.

The forget gate ($f_t$) of the cell at time step $t$ can also be obtained using

$$f_t = \sigma \left( W_{fx} x_t + W_{fh} h_{t-1} + W_{fc} c_{t-1} + b_f \right. \tag{5}$$

where;

$c_{t-1}$ is the previous cell state, $c_t$ is the new cell state, $W_{fx}$, $W_{fh}$, $W_{fc}$ are the incoming links associated with the forget gate having an input bias of $b_f$. Finally, the new cell

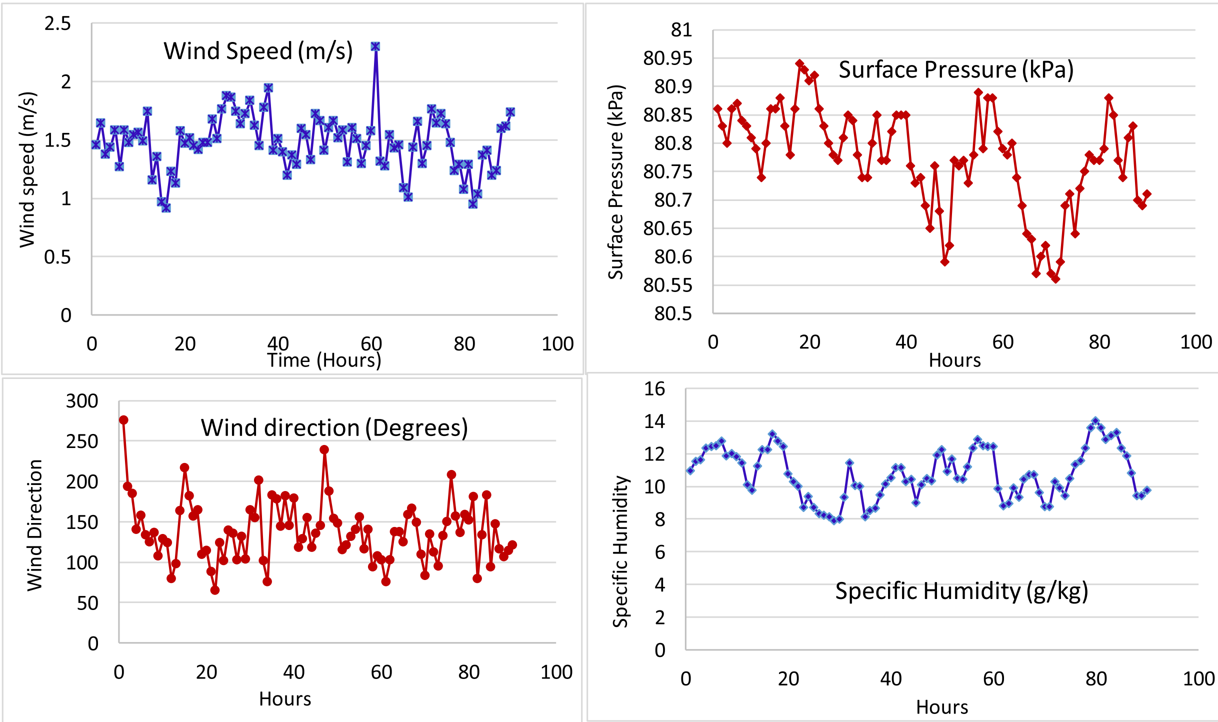


**FIGURE 3.** Wind speed & and direction, surface pressure, and specific humidity of jimma town, february 2023(sources: NASA surface meteorology and solar energy (SSE)).

**TABLE 2.** Historical data of the driving factors of STLF of jimma distribution system.

| Year | Actual Peak load (MW) | Forecasted Peak Load (MW) | GDP (%) | GDP/ Capita | Population | Electricity Consumption (GWh) | Cost of kWh (Birr) | System Losses (%) | Load factor |
|---|---|---|---|---|---|---|---|---|---|
| 2009 | 7.6 | 7.74 | 10 | 392.4 | 193,577 | 4.6 | 0.73 | 19% | 0.71 |
| 2010 | 8.9 | 8.63 | 10.6 | 354.1 | 198,134 | 4.7 | 0.36 | 18% | 0.71 |
| 2011 | 9.8 | 9.76 | 11.4 | 370.3 | 202,798 | 5.1 | 0.5 | 15% | 0.72 |
| 2012 | 10.8 | 11.3 | 8.7 | 491.3 | 207,573 | 5.4 | 0.55 | 14% | 0.72 |
| 2013 | 12.9 | 12.74 | 9.9 | 517.7 | 212,347 | 5.7 | 0.56 | 14% | 0.73 |
| 2014 | 14.35 | 14.26 | 10.3 | 548 | 217,231 | 5.8 | 0.57 | 14% | 0.73 |
| 2015 | 16.27 | 16.18 | 10.2 | 597.3 | 222,227 | 6.9 | 0.59 | 14% | 0.73 |
| 2016 | 18.45 | 18.23 | 6.5 | 647.3 | 227,338 | 8 | 0.61 | 14% | 0.75 |
| 2017 | 20.92 | 20.78 | 7.5 | 697.5 | 232,567 | 11.06 | 0.69 | 14% | 0.75 |
| 2018 | 23.71 | 23.69 | 7.8 | 754.8 | 237,916 | 15.9 | 0.7 | 14% | 0.75 |

state which is a transformed input is given by;

$$c_t = f_t c_{t-1} + i_t \tanh\left(W_{cx} x_t + W_{ch} h_{t-1} + b_c\right) \quad (6)$$

With the above three equations of the LSTM, the output gate of a particular LSTM cell can be calculated using

$$o_t = \sigma\left(W_{ox} x_t + W_{oh} h_{t-1} + W_{oc} c_{t-1} + b_o\right) \quad (7)$$

where, the new hidden state $h_t$ is

$$h_t = o_t \tanh\left(c_t\right) \quad (8)$$

Finally, the output of the cell memory which is the predicted value $y_t$ in this case can be determined from the new hidden states as

$$y_t = W_{yh} h_t + b_y \quad (9)$$

Status update for the created LSTM network of cells can also be determined from the following set of equations

$$c_{tnew} = f_t c_{t-1} + i_t c_t \quad (10)$$

$$h_{tnew} = o_t \tanh\left(c_{tnew}\right) \quad (11)$$

$$W_{ij}^{new} = W_{ij}^{old} + \eta \frac{\partial E}{\partial W_{ij}} \quad (12)$$

where $\eta$ is the training rate of the LSTM network and E is its prediction error given by;

$$E = \sum_{j=1}^{N} \sum_{k=1}^{M} \left(b_{jk} - z_{jk}\right)^2 \quad (13)$$

In this equation, $b_{jk}$ and $z_{jk}$ are the observed and predicted values of the LSTM model output, respectively.

**TABLE 3. Real-time invoices of energy consumption data for jimma distribution system used as source of energy consumption data (actual energy consumption in kWh for march 2022- may 2023).**

| Name of CSCs | Total No. of Invoice Generated | Total INVOICED Amount | Total No. of Invoice Collection | Total Amount of Invoice Collection | Total Amount of Invoice Collection Z fica 08 | No. of Invoice Collection through CashDesk | Invoice Collection Amount through CashDesk | No. of Invoice Collection through Internet | Invoice Collection Amount through Internet | Prepaid Collection | | Performance Efficency in | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | No. | ETB | | ETB | ETB | No. | ETB | No. | ETB | Invoice Collected | ETB | No. Of Invoice | Invoice Amount |
| | **75,684** | **77,646,677.95** | **35,645** | **20,520,492.84** | **1,049,840.46** | **27,923** | **13,370,309.33** | **7,722** | **3,986,583.05** | **9,249** | **2,113,760.00** | **47.10%** | **26.43%** |
| JIMMA DST JIMMA CSC NO.1 | 13,779 | 24,548,128.35 | 6,279 | 7,332,133.96 | 642,943.31 | 5,187 | 4,994,451.09 | 1,092 | 877,339.56 | 3688 | 817,400.00 | 45.57% | 29.87% |
| JIMMA DST JIMMA CSC NO.2 | 10,066 | 14,479,783.68 | 4,568 | 4,927,880.67 | 220,585.28 | 3,475 | 2,906,886.05 | 1,093 | 574,589.34 | 5252 | 1,225,820.00 | 45.38% | 34.03% |
| JIMMA DST ASENDIBO CSC | 4,908 | 6,720,792.93 | 1,831 | 1,069,480.46 | 0.00 | 1,566 | 679,013.61 | 265 | 389,766.85 | 3 | 700.00 | 37.31% | 15.91% |
| JIMMA DST LEMU CSC | 4,273 | 3,512,355.62 | 1,807 | 1,087,531.73 | 0.00 | 145 | 46,697.61 | 1,662 | 1,036,684.12 | 27 | 4,150.00 | 42.29% | 30.96% |
| JIMMA DST TOBA CSC | 3,279 | 1,074,818.35 | 1,921 | 322,130.30 | 0.00 | 1,335 | 265,559.35 | 586 | 52,020.95 | 21 | 4,550.00 | 58.58% | 29.97% |
| JIMMA DST SOKORO CSC | 4,424 | 3,563,506.39 | 2,143 | 748,152.14 | 0.00 | 1,298 | 490,240.83 | 845 | 254,981.31 | 13 | 2,930.00 | 48.44% | 20.99% |
| JIMMA DST AGARO CSC | 11,138 | 5,404,096.98 | 6,198 | 1,972,574.10 | 162,221.51 | 6,115 | 1,673,008.42 | 83 | 119,774.17 | 92 | 17,570.00 | 55.65% | 36.50% |
| JIMMA DST DUMTU CSC | 1,663 | 977,250.15 | 899 | 206,235.68 | 0.00 | 646 | 127,258.26 | 253 | 78,477.42 | 4 | 500.00 | 54.06% | 21.10% |
| JIMMA DST GATIRA CSC | 2,773 | 3,059,523.16 | 831 | 272,564.40 | 0.00 | 575 | 122,775.79 | 256 | 143,988.61 | 24 | 5,800.00 | 29.97% | 8.91% |
| JIMMA DST GERA CSC | 2,992 | 1,368,876.59 | 1,992 | 714,094.74 | 2,251.76 | 1,880 | 675,179.95 | 112 | 35,213.03 | 6 | 1,450.00 | 66.58% | 52.17% |
| JIMMA DST DEDO CSC | 1,017 | 1,036,322.64 | 491 | 273,553.68 | 0.00 | 483 | 260,771.83 | 8 | 7,131.85 | 24 | 5,650.00 | 48.28% | 26.40% |
| JIMMA DST SHEBI CSC | 1,473 | 1,190,335.58 | 691 | 193,134.32 | 50.65 | 408 | 142,392.25 | 283 | 39,191.42 | 10 | 11,500.00 | 46.91% | 16.23% |
| AGRO CSC 2 | 11,913 | 10,454,496.00 | 4,518 | 1,236,664.88 | 15,025.33 | 3,438 | 856,342.10 | 1,080 | 356,747.45 | 41 | 8,550.00 | 37.92% | 11.83% |
| DEMBI CSC | 1,986 | 256,391.53 | 1,476 | 164,361.78 | 6,762.62 | 1,372 | 129,732.19 | 104 | 20,676.97 | 44 | 7,190.00 | 74.32% | 64.11% |

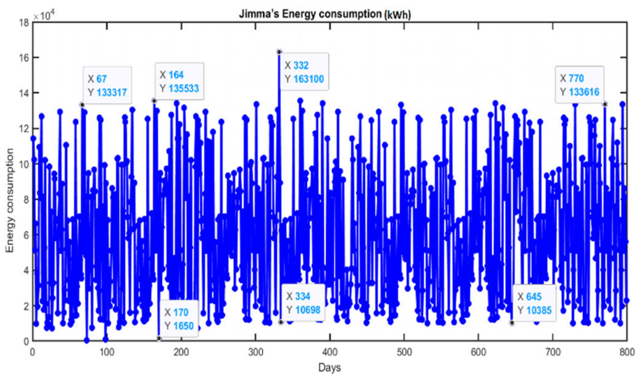


**FIGURE 4.** Jimma town's energy consumption in kWh, march 2022-may 2023.

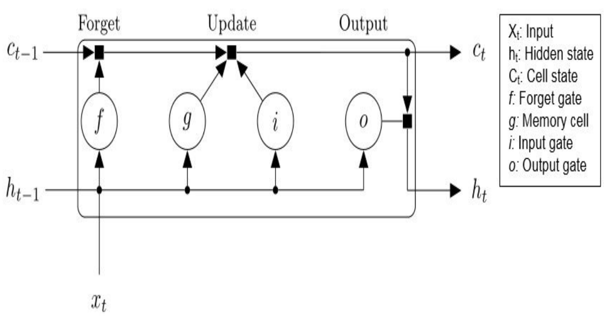


**FIGURE 5.** Long short-term memory (LSTM) cell architecture (source: https://www.mathworks.com/discovery/lstm).

In order to employ LSTM for STLF effectively, LSTM cells are concatenated to form a deep learning model. The number of LSTM cell numbers is determined experimentally based on the complexity of the data to predicted.

A block diagram of the general steps followed in conducting STLF of Jimma city electric power distribution system using EPGA-enhanced LSTM starting from collecting data to the prediction of the load is presented in Figure 6.

The main tasks carried out are data collection, combining the data for weather-dependent STLF using EPGA, training the LSTM model, and load prediction. The ideal solution using EPGA is found through random selection of information exchange by simulating the ability to survive among string structures of the combined equation of the EPGA.

Every time there is a reproduction, a new batch of strong individuals is created from the remains of the old, fittest ones. EPGA for this study is favoured because of its effectiveness in simultaneously solving multiple optimization objectives. EPGA is used to fuse the weather data and historical load, simply to determine the coefficients given in equation (2).

The LSTM weights (as given in equation (4) for a single cell) are determined during the training phase of the model.

$$W = \left[W_{ix}, W_{ih}, W_{fx}, W_{fh}, W_{cx}, W_{ch}, W_{ox}, W_{oh}, W_{yh}\right] \quad (14)$$

Forecasting the hourly electricity load of the next day assists in optimizing the resources and minimizing energy usage. STLF predicts hourly or daily electricity load of one hour to one week which is crucial for resource planning and load balance in power system management. The STLF's key performance indicators are mean squared error (MSE), mean absolute error (MAE) and mean absolute percentage error (MAPE) as given below.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2 \quad (15)$$

$$MAE = \sum_{i=1}^{n}\frac{(y_i - x_i)}{n} \quad (16)$$

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\frac{(y_i - x_i)}{n} \quad (17)$$

In this study, RMSE and MAPE key performance indicators have been used to quantify the prediction performance of LSTM before EPGA enhancement and after EPGA-based

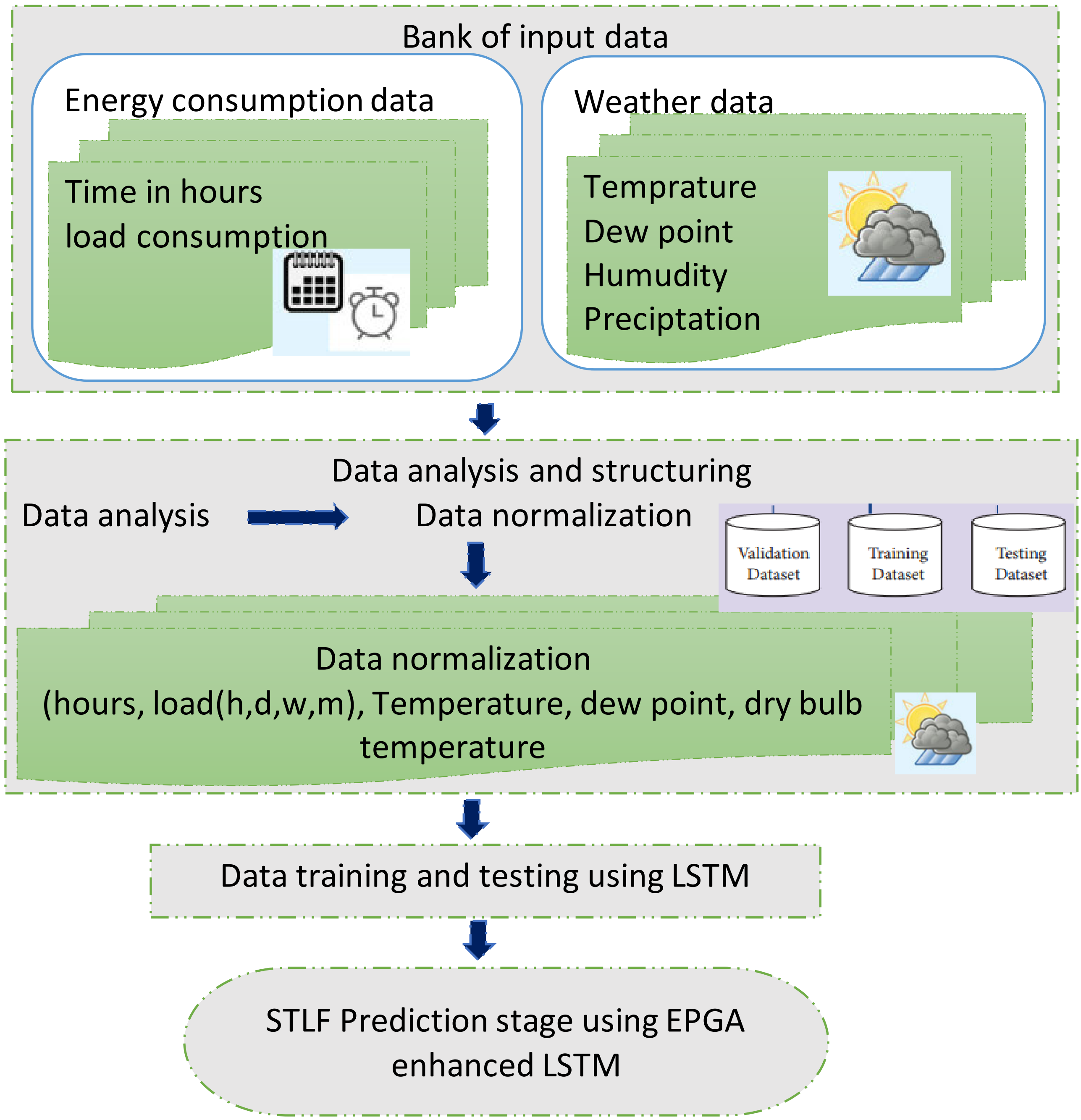


**FIGURE 6.** The block diagram of STLF of electrical power distribution systems using EPGA enhanced LSTM.

enhancements. With the aforementioned models in mind, the proposed EPGA-enhanced LSTM algorithm's flowchart is presented in Figure 7.

## V. RESULTS AND DISCUSSIONS

Both EPGA and LSTM models were trained and tested on data collected pertinent Jimma city electric power distribution system. First the weather data and historical load were fused using EPGA. After the fusion of the data, the LSTM model was trained for STLF on this data. MATLAB software was used to implement the proposed models.

In this work, the LSTM model has a one-dimensional input after data fusion using EPGA. Extensive experiments were done to determine the optimal number of LSTM cells required for STLF of Jimma city electric power distribution system. The model was trained using 710 days of data and tested on 90 days of data as shown in Figure 9.

Using exhaustive search algorithm, we have obtained the optimal number of the LSTM cells to be 500. The training

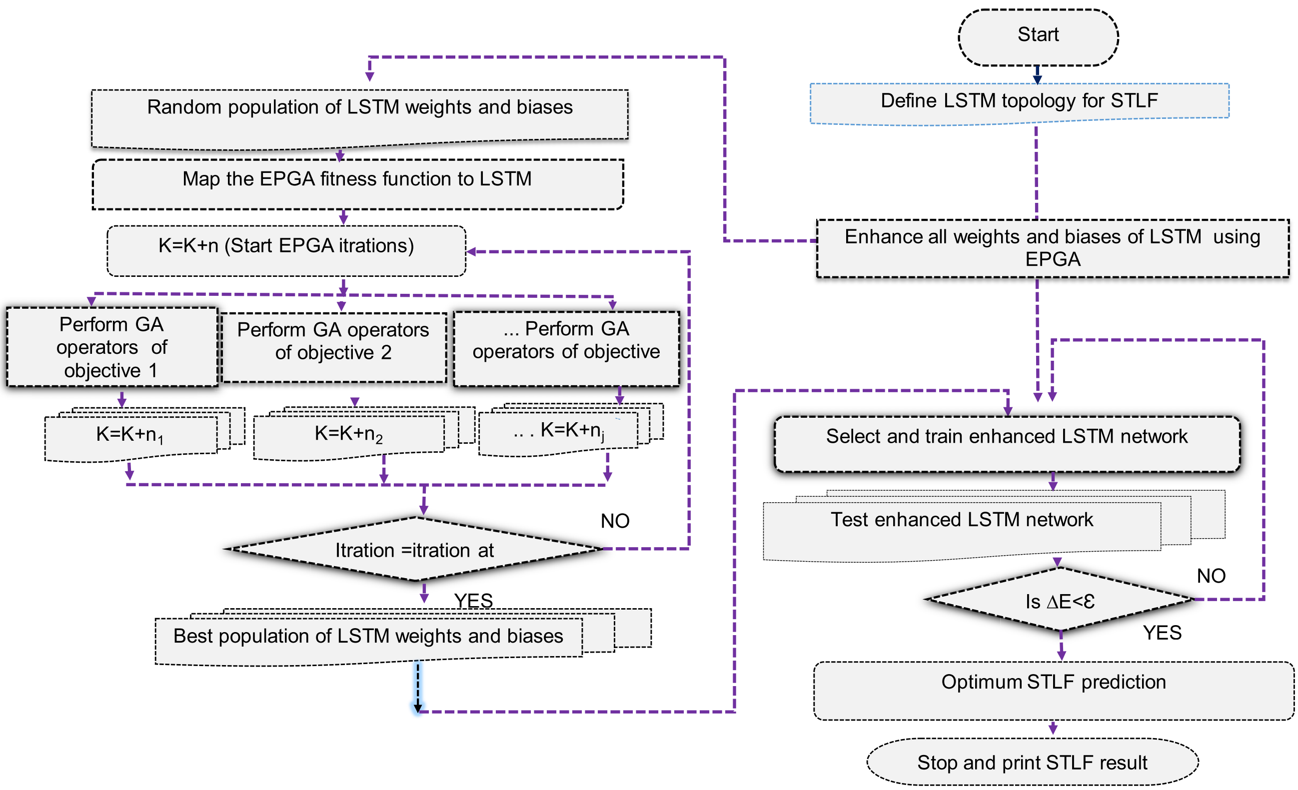


**FIGURE 7.** The flowchart of EPGA enhanced LSTM.

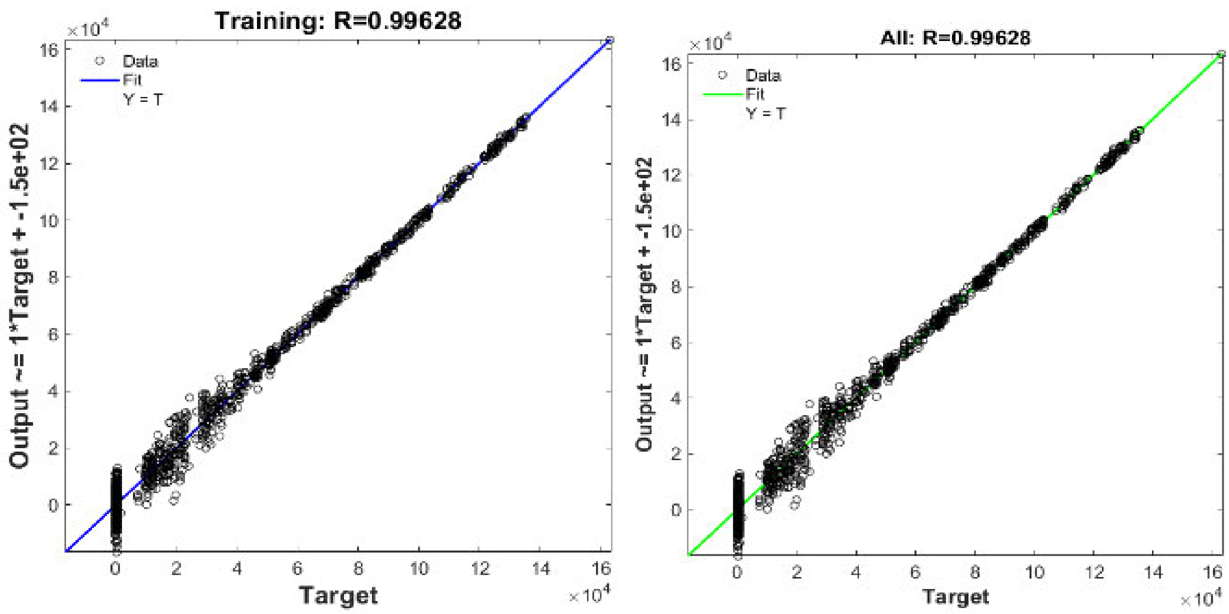


**FIGURE 8.** Regression plot of training the optimal EPGA enhanced LSTM.

process of the proposed LSTM model is shown in Figure 10. As depicted in the figure, the training has saturated at 250 epochs. Training the model beyond that has not led to the improvement in model's prediction accuracy.

After defining the LSTM topology, and mapping the STLF with the loss function of LSTM and with the fitness function of EPGA, all input conditions in the training data were evaluated and trained. Then the enhanced and optimal LSTM is

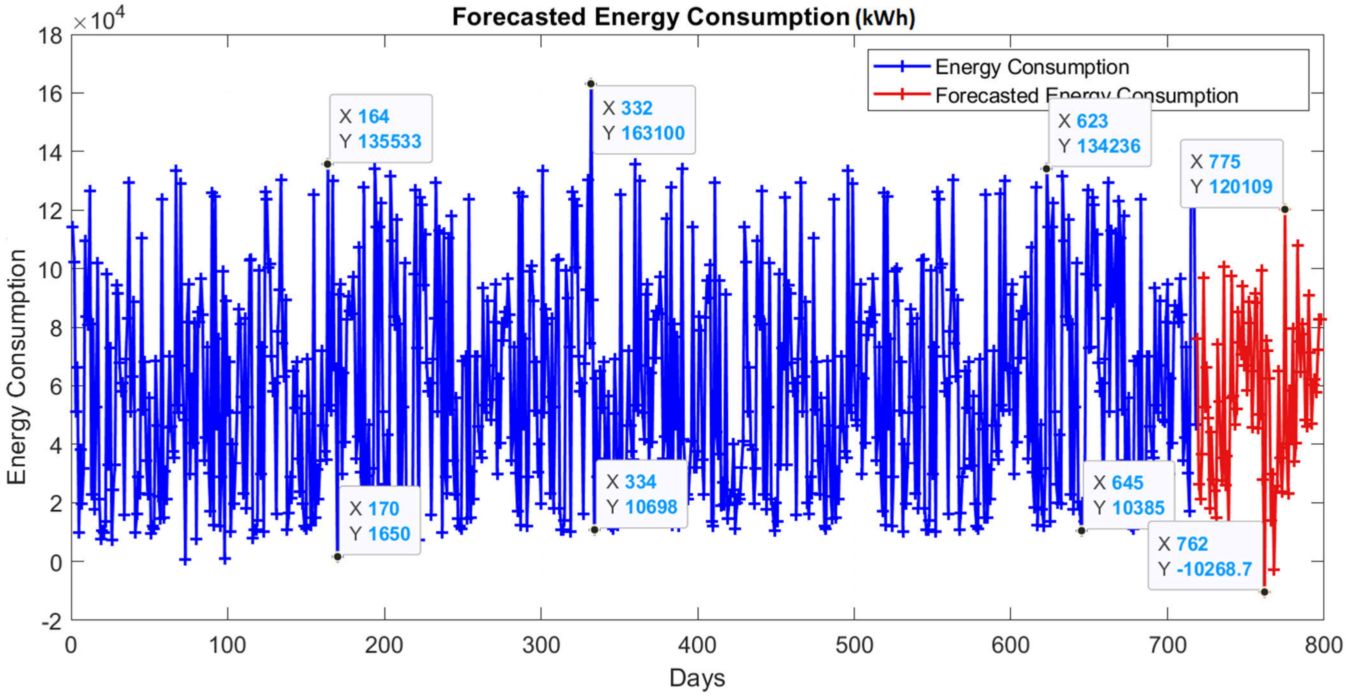


**FIGURE 9.** The forecasted energy consumption of jimma distribution system in kWh, June 2023-Novemebr 2023.

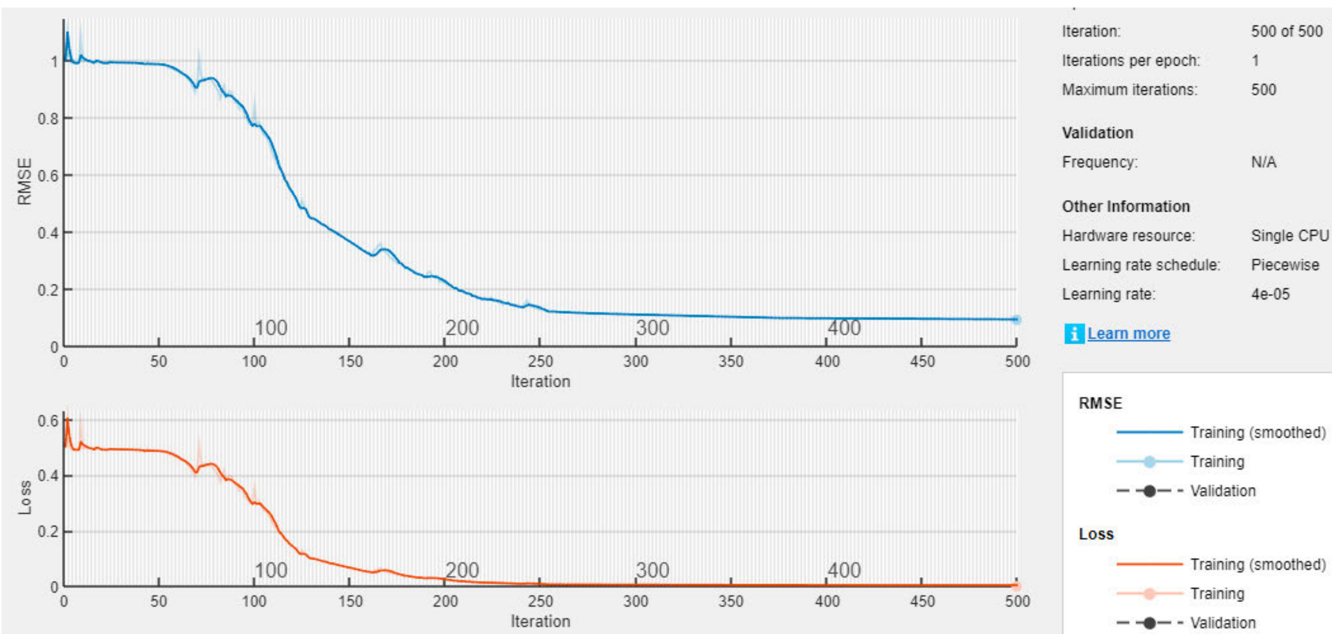


**FIGURE 10.** LSTM training performance in terms of iterations, epochs, RMSE, loss and learning rate of different sub-distribution systems within the Jimma distribution system.

selected and trained to be within the specified tolerance of the entire network. The regression plot of training the optimal LSTM network is presented in Figure 8.

Figure 9 presents the forecasted energy consumption of Jimma distribution system in kWh. In this result, it can be seen that the LSTM has used historical consumption data presented in blue (days 0-710) to predict a day ahead load forecast presented in red (days 710-800). This prediction was made without the employment of any EPGA enhancements STLF is affected by weather, air temperature, and wind speed which are prone to climate change and uncertainty. This leads to uncertain and variable weather profiles. As the

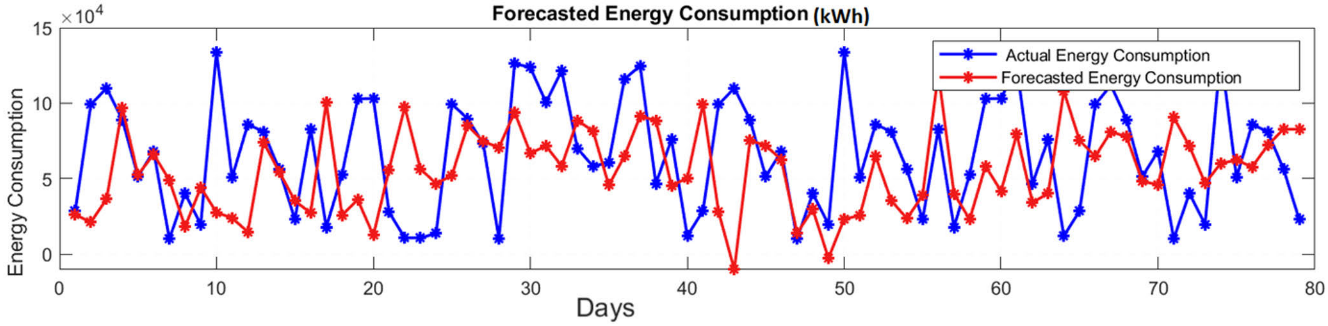


**FIGURE 11.** The predicted energy consumption of Jimma distribution system using LSTM only for June 2023-August 2023.

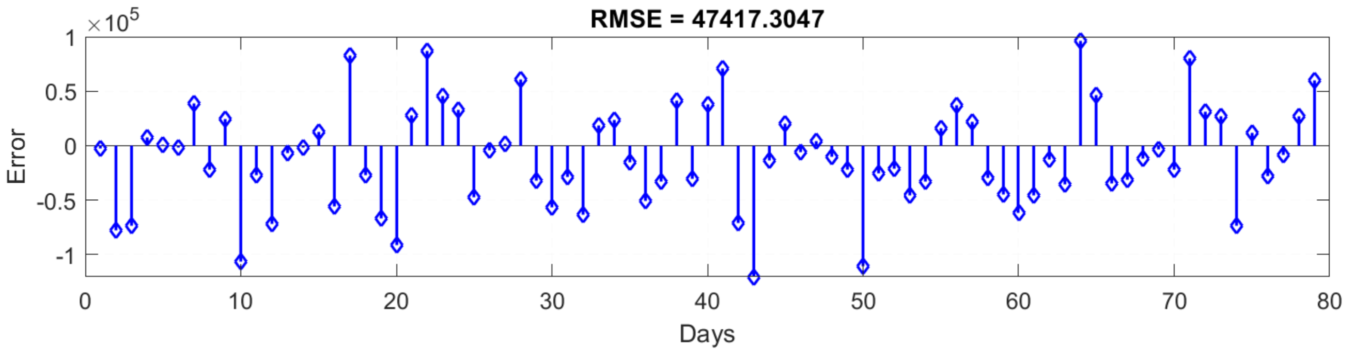


**FIGURE 12.** RMSE of the predicted energy consumption of Jimma distribution system using LSTM only, June 2023-August 2023.

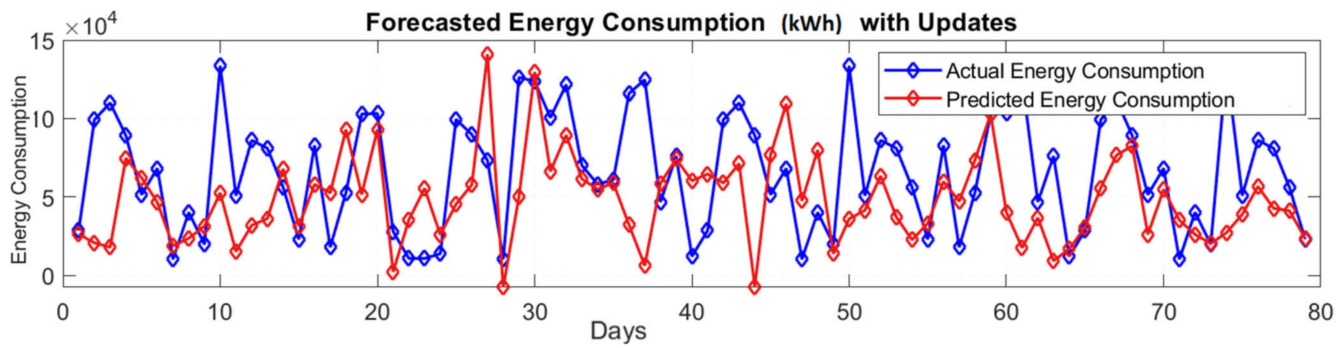


**FIGURE 13.** Forecasted energy consumption of Jimma distribution system using EPGA enhanced LSTM for June 2023-August 2023.

weather is a critical factor that influences forecast accuracy, the weather load model of STLF should be selected to grasp the fluctuations of the load and generation. The behaviours of these factors are mirrored in the demand requirements modelling, though some are affected more than others. Weather-sensitive loads like heating, air conditioning, and ventilating considerably affect small-scale industrial energy sectors.

The proposed STLF model uses the fusion of weather and historical load data, making it suitable to handle variable weather conditions that affect the load demands. As illustration, one can easily see that the proposed model was able to predict the load in a reasonable pattern as shown in Figure 9. As the days ahead increases, the prediction accuracy reduces significantly. Hence, the proposed model is accurate for STLF.

Figure 10 inoculates LSTM training performance in terms of iterations, epochs, RMSE, loss and learning rate. From this figure, it can be easily deduced that the optimal value of the LSTM-based STLF is stabilizing after 215 iterations Figure 11 presents the predicted energy consumption of Jimma distribution system using LSTM only. This result implies that there were no employments of EPGA enhancements.

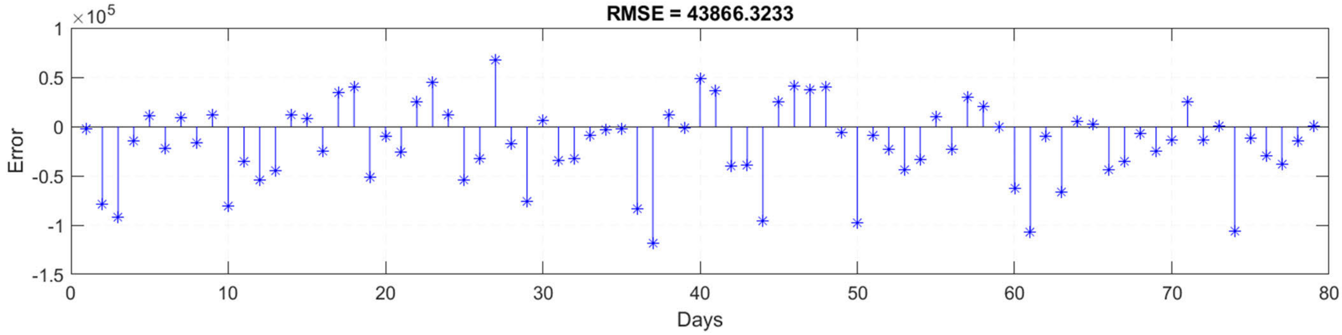


**FIGURE 14.** RSME of Forecasted energy consumption of Jimma distribution system using EPGA enhanced LSTM, June 2023-August 2023.

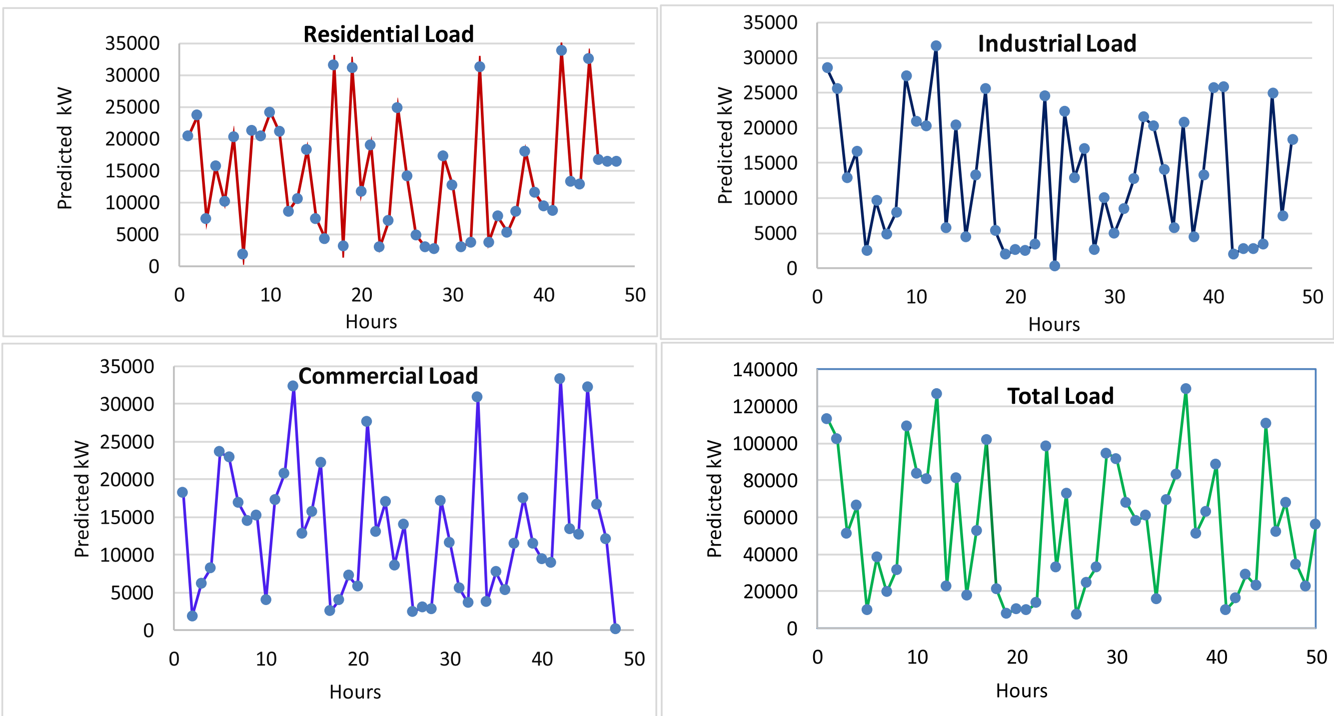


**FIGURE 15.** Sector-wise STLF prediction results of Jimma distribution system for June 2023-July 2023.

With only LSTM, the RMSE of Jimma's STLF is about 47.42 as presented in Figure 12. This result's accuracy should be improved by introducing EPGA enhancements and re-training the input data.

The forecasted energy consumption of Jimma distribution system using EPGA-enhanced LSTM is depicted in Figure 13. As expected, the RMSE of Jimma's STLF using EPGA-enhanced LSTM is found to be 43.87 as shown in Figure 14. In analytical terms, this result is improved by 7.486% of its original prediction accuracy.

The sector-wise STLF predictions of Jimma distribution system that take into account historical energy usage data, weather and economic factors are presented in Figure 15. The robustness of the EPGA-enhanced LSTM-based STLF in terms of MAPE is compared with other two hybrid methods from the literature [20], [40] and one LSTM method of our study as presented in Table 4.

In addition to the MAPE presented above, this table depicts the employment of LSTM, GCN-LSTM [40], and PDLSTM-CNN [20] of five sample consumption data against EPGA-enhanced LSTM to indicate the performance of the solution method in accordance with the proposed mathematical model. Fortunately, Jimma's climate is relatively cool tropical monsoon with no significant difference throughout the year. This signifies there is no significant effect of the weather model bestowed on the STLF.

However, this model helps in grasping the effect of unaccounted-for distributed generators such as solar home systems (SHSs) on the historical energy consumption data of Jimma. The weather model of STLF using EPGA-enhanced

**TABLE 4. MAPE of EPGA enhanced LSTM-based STLF for five sample loads.**

| Sample Load | Actual Consumption (kW) | LSTM prediction (MAPE) | GCN-LSTM prediction (MAPE)[40] | PDLSTM-CNN prediction (MAPE)[20] | Proposed EPGA-LSTM prediction (MAPE) |
|---|---|---|---|---|---|
| 1 | 114,000.00 | 5.62 | 7.46 | 8.98 | 4.93 |
| 2 | 102,000.00 | 3.41 | 4.81 | 5.26 | 2.93 |
| 3 | 51,200.00 | 1.04 | 1.19 | 3.07 | 0.09 |
| 4 | 66,300.00 | 0.07 | 0.89 | 1.83 | 0.00 |
| 5 | 9,830.00 | 0.00 | 0.14 | 0.92 | 0.00 |

LSTM is critically important for case studies with significant weather changes and impacts as it helps to improve the prediction accuracy of STLF from both the generation side and the load/demand side.

## VI. CONCLUSION

Balancing demand and supply in the economics of power systems planning and operation has been a moving target for decades. This moving target gets complex and cumbersome with the introduction of intermittent renewables and variable needs for electricity. One power system operation task that helps in addressing these challenges is Load Forecasting (LF). STLF ranging from day ahead hours to weeks is a computationally difficult optimization problem.

However, STLF is vital for an electric industry in a deregulated energy market. In this work, STLF uses enhanced Deep Neural Networks (DNNs) to quantify the uncertainty in the future load of Jimma city electric power distribution system by developing accurate, reliable and stable model that specify the relationship between load and its key drivers presented. As a result:

- An enhanced, continuous, multilayered Recurrent Neural Network (RNN) called Long Short-Term Memory (LSTM) with Efficient and Parallel Genetic Algorithm's (EPGA) arithmetic operations inspired enhancements is introduced and used for STLF of Jimma distribution system.
- In analytical terms, this result is improved by 7.486% of its original prediction accuracy. MAPE of five sample loads considered in the test of the robustness of the EPGA-enhanced LSTM prediction methodology also backed the performance of the proposed model.

In summary, STLF of power distribution systems with high penetration of renewable energy systems and distributed generations such as SHS using EPGA enhanced LSTM is promisingly efficient and effective.

## VII. FUTURE WORK AND RECOMMENDATIONS

The foresight of EPGA-enhanced LSTM-based STLF is imperative in every aspect of power system operation and planning. In Jimma city, the case study considered, there are no significant effects of the weather model bestowed on the STLF. However, this model helps in grasping the effect of unaccounted-for distributed generators such as SHSs on the historical energy consumption data of Jimma.

- The weather model of STLF using EPGA-enhanced LSTM is critically important for case studies with significant weather changes and impacts as it helps to improve the prediction accuracy of STLF from both the generation side and the load/demand side.
- As future power system structures will comprise smart grids, energy prosumers, demand response, electric vehicles, big data-driven models, data automation meters, and big data-based augmented control systems; it is crucial to update the weather load model of STLF using hybridization or enhancements of deeply learning methodologies.
- For this reason, special attention should be given to STLF using EPGA-enhanced LSTM that considers weather load model, big-data-driven modelling and enhanced DNNs with hybridization features from the perspective of distributed generators, microgrids, energy storage mechanisms and demand response.

As the enhanced and hybridized computational algorithms strengthen and advance, timely STLF models that cope with the evolution of deep learning methods are thus recommended. In compliance with these recommendations, hybridization and further enhancement of the EPGA-enhanced LSTM-based STLF for large power systems, power pools and area controls are proposed as future work.

## REFERENCES

[1] A. Alrasheedi and A. Almalaq, "Hybrid deep learning applied on Saudi smart grids for short-term load forecasting," *Mathematics*, vol. 10, no. 15, p. 2666, Jul. 2022.

[2] W.-C. Yeh, M.-F. He, C.-L. Huang, S.-Y. Tan, X. Zhang, Y. Huang, and L. Li, "New genetic algorithm for economic dispatch of stand-alone three-modular microgrid in DongAo island," *Appl. Energy*, vol. 263, Apr. 2020, Art. no. 114508, doi: 10.1016/j.apenergy.2020.114508.

[3] S. Tsegaye, F. Shewarega, and G. Bekele, "Artificial intelligence based security constrained economic dispatch of Ethiopian renewable energy systems: A comparative study," in *Proc. Int. Conf. Adv. Sci. Technol.*, in Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol. 412. Cham, Switzerland: Springer, 2022, pp. 522–542, doi: 10.1007/978-3-030-93712-6_35.

[4] S. Tsegaye and G. Bekele, "Optimal generation dispatch of Ethiopian power system using hybrid genetic algorithm-hopfield neural network," *EAI Endorsed Trans. Energy Web*, vol. 9, no. 37, pp. 1–15, 2022, doi: 10.4108/eai.13-8-2021.170673.

[5] M. Abdel-Basset, H. Hawash, K. Sallam, S. S. Askar, and M. Abouhawwash, "STLF-Net: Two-stream deep network for short-term load forecasting in residential buildings," *J. King Saud Univ., Comput. Inf. Sci.*, vol. 34, no. 7, pp. 4296–4311, Jul. 2022, doi: 10.1016/j.jksuci.2022.04.016.

[6] P. Roy and A. Chakrabarti, "Modified shuffled frog leaping algorithm for solving economic load dispatch problem," *Energy Power Eng.*, vol. 3, no. 4, pp. 551–556, 2011, doi: 10.4236/epe.2011.34068.

[7] M. L. Abdulrahman, K. M. Ibrahim, A. Y. Gital, F. U. Zambuk, B. Ja'afaru, Z. I. Yakubu, and A. Ibrahim, "A review on deep learning with focus on deep recurrent neural network for electricity forecasting in residential building," *Proc. Comput. Sci.*, vol. 193, pp. 141–154, Jan. 2021, doi: 10.1016/j.procs.2021.10.014.

[8] S. Arastehfar, M. Matinkia, and M. R. Jabbarpour, "Short-term residential load forecasting using graph convolutional recurrent neural networks," *Eng. Appl. Artif. Intell.*, vol. 116, Nov. 2022, Art. no. 105358, doi: 10.1016/j.engappai.2022.105358.

[9] F. Mehmood Butt, L. Hussain, A. Mahmood, and K. J. Lone, "Artificial intelligence based accurately load forecasting system to forecast short and medium-term load demands," *Math. Biosci. Eng.*, vol. 18, no. 1, pp. 400–425, 2021, doi: 10.3934/mbe.2021022.

[10] P.-H. Kuo and C.-J. Huang, "A high precision artificial neural networks model for short-term energy load forecasting," *Energies*, vol. 11, no. 1, p. 213, Jan. 2018, doi: 10.3390/en11010213.

[11] A. Mansouri, A. H. Abolmasoumi, and A. A. Ghadimi, "Weather sensitive short term load forecasting using dynamic mode decomposition with control," *Electr. Power Syst. Res.*, vol. 221, Aug. 2023, Art. no. 109387, doi: 10.1016/j.epsr.2023.109387.

[12] Y. Xuan, W. Si, J. Zhu, Z. Sun, J. Zhao, M. Xu, and S. Xu, "Multi-model fusion short-term load forecasting based on random forest feature selection and hybrid neural network," *IEEE Access*, vol. 9, pp. 69002–69009, 2021, doi: 10.1109/ACCESS.2021.3051337.

[13] J. Wang, X. Chen, F. Zhang, F. Chen, and Y. Xin, "Building load forecasting using deep neural network with efficient feature fusion," *J. Mod. Power Syst. Clean Energy*, vol. 9, no. 1, pp. 160–169, Jan. 2021, doi: 10.35833/MPCE.2020.000321.

[14] B. U. Islam and S. F. Ahmed, "Short-term electrical load demand forecasting based on LSTM and RNN deep neural networks," *Math. Problems Eng.*, vol. 2022, Jul. 2022, Art. no. 2316474.

[15] P. Liu, P. Zheng, and Z. Chen, "Deep learning with stacked denoising autoencoder for short-term electric load forecasting," *Energies*, vol. 12, no. 12, p. 2445, Jun. 2019, doi: 10.3390/en12122445.

[16] K. Wang, X. Qi, and H. Liu, "A comparison of day-ahead photovoltaic power forecasting models based on deep learning neural network," *Appl. Energy*, vol. 251, Oct. 2019, Art. no. 113315, doi: 10.1016/j.apenergy.2019.113315.

[17] D. L. Marino, K. Amarasinghe, and M. Manic, "Building energy load forecasting using deep neural networks," in *Proc. 42nd Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2016, pp. 7046–7051, doi: 10.1109/IECON.2016.7793413.

[18] C. Fan, J. Wang, W. Gang, and S. Li, "Assessment of deep recurrent neural network-based strategies for short-term building energy predictions," *Appl. Energy*, vol. 236, pp. 700–710, Feb. 2019, doi: 10.1016/j.apenergy.2018.12.004.

[19] M. Cai, M. Pipattanasomporn, and S. Rahman, "Day-ahead building-level load forecasts using deep learning vs. traditional time-series techniques," *Appl. Energy*, vol. 236, pp. 1078–1088, Feb. 2019, doi: 10.1016/j.apenergy.2018.12.042.

[20] B. Farsi, M. Amayri, N. Bouguila, and U. Eicker, "On short-term load forecasting using machine learning techniques and a novel parallel deep LSTM-CNN approach," *IEEE Access*, vol. 9, pp. 31191–31212, 2021, doi: 10.1109/ACCESS.2021.3060290.

[21] O. Jogunola, B. Adebisi, K. V. Hoang, Y. Tsado, S. I. Popoola, M. Hammoudeh, and R. Nawaz, "CBLSTM-AE: A hybrid deep learning framework for predicting energy consumption," *Energies*, vol. 15, no. 3, p. 810, Jan. 2022, doi: 10.3390/en15030810.

[22] M. Sajjad, Z. A. Khan, A. Ullah, T. Hussain, W. Ullah, M. Y. Lee, and S. W. Baik, "A novel CNN-GRU-based hybrid approach for short-term residential load forecasting," *IEEE Access*, vol. 8, pp. 143759–143768, 2020, doi: 10.1109/ACCESS.2020.3009537.

[23] R. Wan, S. Mei, J. Wang, M. Liu, and F. Yang, "Multivariate temporal convolutional network: A deep neural networks approach for multivariate time series forecasting," *Electronics*, vol. 8, no. 8, p. 876, Aug. 2019, doi: 10.3390/electronics8080876.

[24] C.-W. Wang, M.-A. Khalil, and N. Firdi, "A survey on deep learning for precision oncology," *Diagnostics*, vol. 12, no. 6, p. 1489, Jun. 2022, doi: 10.3390/diagnostics12061489.

[25] A. Moradzadeh, S. Zakeri, M. Shoaran, B. Mohammadi-Ivatloo, and F. Mohammadi, "Short-term load forecasting of microgrid via hybrid support vector regression and long short-term memory algorithms," *Sustainability*, vol. 12, no. 17, p. 7076, Aug. 2020, doi: 10.3390/su12177076.

[26] C. Sekhar and R. Dahiya, "Robust framework based on hybrid deep learning approach for short term load forecasting of building electricity demand," *Energy*, vol. 268, Apr. 2023, Art. no. 126660, doi: 10.1016/j.energy.2023.126660.

[27] A. Mellit, A. M. Pavan, and V. Lughi, "Deep learning neural networks for short-term photovoltaic power forecasting," *Renew. Energy*, vol. 172, pp. 276–288, Jul. 2021, doi: 10.1016/j.renene.2021.02.166.

[28] V. Y. Kondaiah, B. Saravanan, P. Sanjeevikumar, and B. Khan, "A review on short-term load forecasting models for micro-grid application," *J. Eng.*, vol. 2022, no. 7, pp. 665–689, Jul. 2022, doi: 10.1049/tje2.12151.

[29] S. Zhang, R. Chen, J. Cao, and J. Tan, "A CNN and LSTM-based multi-task learning architecture for short and medium-term electricity load forecasting," *Electr. Power Syst. Res.*, vol. 222, Sep. 2023, Art. no. 109507, doi: 10.1016/j.epsr.2023.109507.

[30] H. H. Goh, B. He, H. Liu, D. Zhang, W. Dai, T. A. Kurniawan, and K. C. Goh, "Multi-convolution feature extraction and recurrent neural network dependent model for short-term load forecasting," *IEEE Access*, vol. 9, pp. 118528–118540, 2021, doi: 10.1109/ACCESS.2021.3107954.

[31] C. Tian, J. Ma, C. Zhang, and P. Zhan, "A deep neural network model for short-term load forecast based on long short-term memory network and convolutional neural network," *Energies*, vol. 11, no. 12, p. 3493, Dec. 2018, doi: 10.3390/en11123493.

[32] Y. Xie, Y. Ueda, and M. Sugiyama, "A two-stage short-term load forecasting method using long short-term memory and multilayer perceptron," *Energies*, vol. 14, no. 18, p. 5873, Sep. 2021.

[33] H. Bian, Q. Wang, G. Xu, and X. Zhao, "Load forecasting of hybrid deep learning model considering accumulated temperature effect," *Energy Rep.*, vol. 8, pp. 205–215, Apr. 2022, doi: 10.1016/j.egyr.2021.11.082.

[34] F. Zhou, H. Zhou, Z. Li, and K. Zhao, "Multi-step ahead short-term electricity load forecasting using VMD-TCN and error correction strategy," *Energies*, vol. 15, no. 15, p. 5375, Jul. 2022.

[35] L. Gong, "Application of CNN-LSTM based hybrid neural network in power load forecasting," Res. Square, 2022, pp. 1–18.

[36] S. Yi, H. Liu, T. Chen, J. Zhang, and Y. Fan, "A deep LSTM-CNN based on self-attention mechanism with input data reduction for short-term load forecasting," *IET Gener., Transmiss. Distrib.*, vol. 17, no. 7, pp. 1538–1552, Apr. 2023, doi: 10.1049/gtd2.12763.

[37] S. Atef, K. Nakata, and A. B. Eltawil, "A deep bi-directional long-short term memory neural network-based methodology to enhance short-term electricity load forecasting for residential applications," *Comput. Ind. Eng.*, vol. 170, Aug. 2022, Art. no. 108364, doi: 10.1016/j.cie.2022.108364.

[38] K. S. Sudheera, R. Swetha, R. Tejaswini, M. V. Meena, and A. G. Kumar, "Residential load forecasting based on deep neural network," in *Proc. 4th Int. Conf. Innov. Trends Inf. Technol. (ICITIIT)*, Feb. 2023, pp. 1–6, doi: 10.1109/ICITIIT57246.2023.10068706.

[39] S. S. Subbiah and J. Chinnappan, "Deep learning based short term load forecasting with hybrid feature selection," *Electr. Power Syst. Res.*, vol. 210, Sep. 2022, Art. no. 108065, doi: 10.1016/j.epsr.2022.108065.

[40] H. Chen, M. Zhu, X. Hu, J. Wang, Y. Sun, and J. Yang, "Research on short-term load forecasting of new-type power system based on GCN-LSTM considering multiple influencing factors," *Energy Rep.*, vol. 9, pp. 1022–1031, Oct. 2023, doi: 10.1016/j.egyr.2023.05.048.

[41] X. Long, X. Gong, B. Zhang, and H. Zhou, "Deep learning based data prefetching in CPU-GPU unified virtual memory," *J. Parallel Distrib. Comput.*, vol. 174, pp. 19–31, Apr. 2023, doi: 10.1016/j.jpdc.2022.12.004.

[42] P.-P. Phyo and C. Jeenanunta, "Advanced ML-based ensemble and deep learning models for short-term load forecasting: Comparative analysis using feature engineering," *Appl. Sci.*, vol. 12, no. 10, p. 4882, May 2022, doi: 10.3390/app12104882.

[43] M. Alhussein, K. Aurangzeb, and S. I. Haider, "Hybrid CNN-LSTM model for short-term individual household load forecasting," *IEEE Access*, vol. 8, pp. 180544–180557, 2020, doi: 10.1109/ACCESS.2020.3028281.

[44] M.-H. Pham, M.-N. Nguyen, and Y.-K. Wu, "A novel short-term load forecasting method by combining the deep learning with singular spectrum analysis," *IEEE Access*, vol. 9, pp. 73736–73746, 2021, doi: 10.1109/ACCESS.2021.3078900.

[45] R. Jiao, T. Zhang, Y. Jiang, and H. He, "Short-term non-residential load forecasting based on multiple sequences LSTM recurrent neural network," *IEEE Access*, vol. 6, pp. 59438–59448, 2018, doi: 10.1109/ACCESS.2018.2873712.

[46] X. Luo and L. O. Oyedele, "A self-adaptive deep learning model for building electricity load prediction with moving horizon," *Mach. Learn. Appl.*, vol. 7, Mar. 2022, Art. no. 100257, doi: 10.1016/j.mlwa.2022.100257.
[47] M. Khan, M. Ishaq, and S. Kwon, "Short-term energy forecasting framework using an ensemble deep learning approach," *IEEE Access*, vol. 9, pp. 94262–94271, 2021, doi: 10.1109/ACCESS.2021.3093053.
[48] Y. Pan, J. Zheng, Y. Yang, R. Zhu, C. Zhou, and T. Shi, "An electricity load forecasting approach combining DBN-based deep neural network and NAR model for the integrated energy systems," in *Proc. IEEE Int. Conf. Big Data Smart Comput. (BigComp)*, Mar. 2019, no. 1, pp. 1–4, doi: 10.1109/BIGCOMP.2019.8679194.
[49] S. M. A. Haque and A. Islam, "Artificial neural network-based short-term load forecasting for Mymensingh area of Bangladesh," *Int. J. Elect. Comput. Eng.*, vol. 15, no. 3, pp. 99–103, 2021.
[50] X. J. Luo, L. O. Oyedele, A. O. Ajayi, O. O. Akinade, H. A. Owolabi, and A. Ahmed, "Feature extraction and genetic algorithm enhanced adaptive deep neural network for energy consumption prediction in buildings," *Renew. Sustain. Energy Rev.*, vol. 131, Oct. 2020, Art. no. 109980, doi: 10.1016/j.rser.2020.109980.
[51] X. J. Luo, L. O. Oyedele, A. O. Ajayi, O. O. Akinade, J. M. D. Delgado, H. A. Owolabi, and A. Ahmed, "Genetic algorithm-determined deep feedforward neural network architecture for predicting electricity consumption in real buildings," *Energy AI*, vol. 2, Nov. 2020, Art. no. 100015, doi: 10.1016/j.egyai.2020.100015.
[52] S. Frank, I. Steponavice, and S. Rebennack, "Optimal power flow: A bibliographic survey II: Non-deterministic and hybrid methods," *Energy Syst.*, vol. 3, no. 3, pp. 259–289, Sep. 2012, doi: 10.1007/s12667-012-0057-x.
[53] E. Balraj, T. Pugalendhi, M. Sureshkumar, and K. Vijayasarathi, "Optimized LSTM model for electric load forecasting using deep learning with genetic algorithm," in *Proc. 7th Int. Conf. Intell. Comput. Control Syst. (ICICCS)*, May 2023, pp. 301–304, doi: 10.1109/ICICCS56967.2023.10142580.
[54] F. Shahid, A. Zameer, and M. Muneeb, "A novel genetic LSTM model for wind power forecast," *Energy*, vol. 223, May 2021, Art. no. 120069, doi: 10.1016/j.energy.2021.120069.
[55] H. Chung and K.-S. Shin, "Genetic algorithm-optimized long short-term memory network for stock market prediction," *Sustainability*, vol. 10, no. 10, p. 3765, Oct. 2018, doi: 10.3390/su10103765.
[56] A. S. Mahajan and A. Shrivastav, "Short term load forecasting based on regression models," in *Proc. Int. Conf. Adv. Technol. (ICONAT)*, Jan. 2023, pp. 1–8, doi: 10.1109/ICONAT57137.2023.10080359.
[57] N. Pradeep and G. S. Sheshadri, "Convolution neural network based short-term load forecasting," *Int. J. Creative Res. Thoughts*, vol. 11, no. 3, pp. 413–418, 2023.
[58] X. J. Luo and L. O. Oyedele, "Forecasting building energy consumption: Adaptive long-short term memory neural networks driven by genetic algorithm," *Adv. Eng. Informat.*, vol. 50, Oct. 2021, Art. no. 101357, doi: 10.1016/j.aei.2021.101357.
[59] D. Wei, J. Wang, K. Ni, and G. Tang, "Research and application of a novel hybrid model based on a deep neural network combined with fuzzy time series for energy forecasting," *Energies*, vol. 12, no. 18, p. 3588, Sep. 2019, doi: 10.3390/en12183588.
[60] L. Han, Y. Peng, Y. Li, B. Yong, Q. Zhou, and L. Shu, "Enhanced deep networks for short-term and medium-term load forecasting," *IEEE Access*, vol. 7, pp. 4045–4055, 2019, doi: 10.1109/ACCESS.2018.2888978.
[61] S. M. Naseem, "Heuristic algorithm based home energy management system in smart grid," M.S. thesis, Dept. Elect. Eng., COMSATS Inst. Inf. Technol. Islamabad, Islamabad, Pakistan, Jan. 2017, doi: 10.13140/RG.2.2.11317.47840.
[62] S. Motepe, A. N. Hasan, and R. Stopforth, "Improving load forecasting process for a power distribution network using hybrid AI and deep learning algorithms," *IEEE Access*, vol. 7, pp. 82584–82598, 2019, doi: 10.1109/ACCESS.2019.2923796.
[63] M. Zulfiqar and M. B. Rasheed, "Short-term load forecasting using long short term memory optimized by genetic algorithm," in *Proc. IEEE Sustain. Power Energy Conf. (iSPEC)*, Dec. 2022, pp. 1–5, doi: 10.1109/iSPEC54162.2022.10033074.
[64] J. Lu, Q. Zhang, Z. Yang, and M. Tu, "A hybrid model based on convolutional neural network and long short-term memory for short-term load forecasting," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Aug. 2019, pp. 1–5, doi: 10.1109/PESGM40551.2019.8973549.
[65] Z. Liao, H. Pan, X. Huang, R. Mo, X. Fan, H. Chen, L. Liu, and Y. Li, "Short-term load forecasting with dense average network," *Expert Syst. Appl.*, vol. 186, Dec. 2021, Art. no. 115748, doi: 10.1016/j.eswa.2021.115748.
[66] J. Sun, M. Gong, Y. Zhao, C. Han, L. Jing, and P. Yang, "A hybrid deep reinforcement learning ensemble optimization model for heat load energy-saving prediction," *J. Building Eng.*, vol. 58, Oct. 2022, Art. no. 105031, doi: 10.1016/j.jobe.2022.105031.
[67] M. R. Habibi, S. Golestan, J. M. Guerrero, and J. C. Vasquez, "Deep learning for forecasting-based applications in cyber–physical microgrids: Recent advances and future directions," *Electronics*, vol. 12, no. 7, p. 1685, Apr. 2023, doi: 10.3390/electronics12071685.
[68] A. Berhanu, "The Ethiopian extension and the farmer: A view from the farm," in *Proc. 16th Int. Conf. Ethiopian Stud.*, 2009, pp. 751–760.
[69] M. A. Tolcha and G. S. Tibba, "Utilization of waste polystyrene material in local foundry technology for manufacturing complex shapes," *Int. J. Eng. Res.*, vol. 3, no. 3, pp. 1975–1979, Mar. 2014.
[70] A. G. Rameshrao, E. Koley, and S. Ghosh, "A LSTM-based approach for detection of high impedance faults in hybrid microgrid with immunity against weather intermittency and N-1 contingency," *Renew. Energy*, vol. 198, pp. 75–90, Oct. 2022, doi: 10.1016/j.renene.2022.08.028.
[71] R. Chinnaraji and P. Ragupathy, "Accurate electricity consumption prediction using enhanced long short-term memory," *IET Commun.*, vol. 16, no. 8, pp. 830–844, May 2022, doi: 10.1049/cmu2.12384.
[72] D. Kaur, S. N. Islam, M. A. Mahmud, M. E. Haque, and Z. Y. Dong, "Energy forecasting in smart grid systems: Recent advancements in probabilistic deep learning," *IET Gener., Transmiss. Distrib.*, vol. 16, no. 22, pp. 4461–4479, Nov. 2022, doi: 10.1049/gtd2.12603.
[73] M. Li, H. Tian, Q. Chen, M. Zhou, and G. Li, "A hybrid prediction method for short-term load based on temporal convolutional networks and attentional mechanisms," *IET Gener., Transmiss. Distrib.*, vol. 18, no. 5, pp. 885–898, Mar. 2024, doi: 10.1049/gtd2.12798.
[74] S. Hosein and P. Hosein, "Load forecasting using deep neural networks," in *Proc. IEEE Power Energy Soc. Innov. Smart Grid Technol. Conf. (ISGT)*, Apr. 2017, pp. 1–5, doi: 10.1109/ISGT.2017.8085971.
[75] L. Zhou, C. Zhao, N. Liu, X. Yao, and Z. Cheng, "Improved LSTM-based deep learning model for COVID-19 prediction using optimized approach," *Eng. Appl. Artif. Intell.*, vol. 122, Jun. 2023, Art. no. 106157, doi: 10.1016/j.engappai.2023.106157.

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